UMHackathon 2026

AI SPENDING DECISION ASSISTANT

STRATATOUILLE

“STOP GUESSING, START SPENDING SMARTER”

THE PROBLEM

Students and young adults face three barriers to smart spending:

1. Limited analytical capabilities
2. Reliance on manual/intuition-based processes
3. Fragmented data

CONSEQUENCE

- Overspending
- Unexpected shortfalls
- Financial stress

TARGET USER GROUP

Primary users: University students (18-25 years old)

USER NEEDS

INSTANT

Record expenses in less than 10 seconds

PREVENTIVE

Know if they can purchase *before* buying

HUMAN-READABLE

Get plain-language advice, not spreadsheets

SOLUTION OVERVIEW

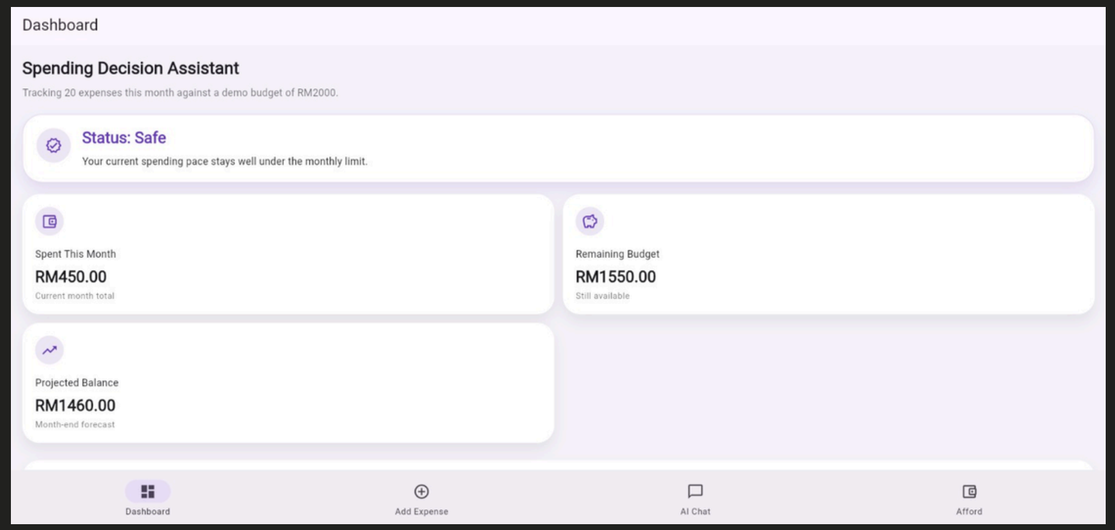
AI Spending Decision Assistant

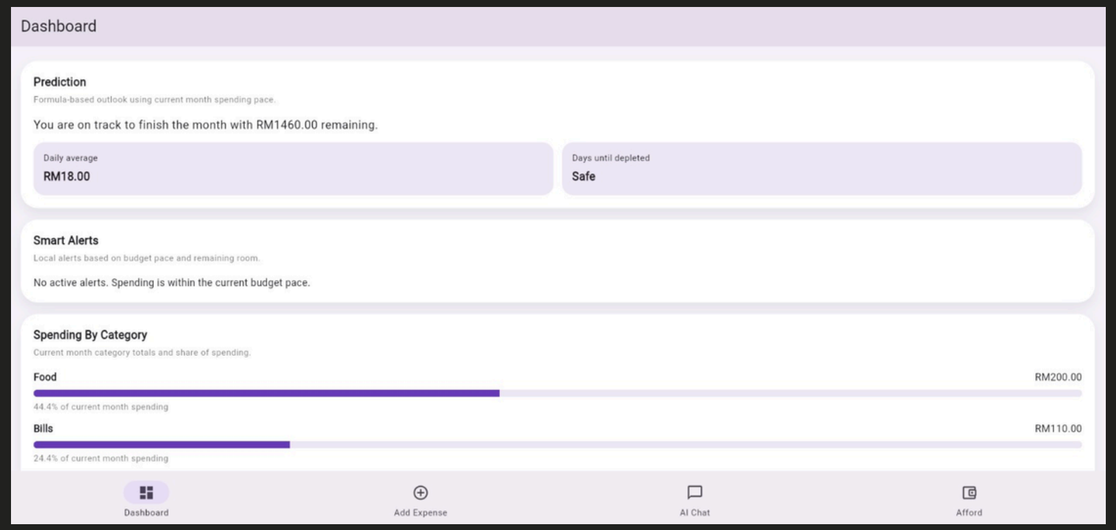
A Flutter + Firebase expense tracker with Z.AI GLM-powered intelligence

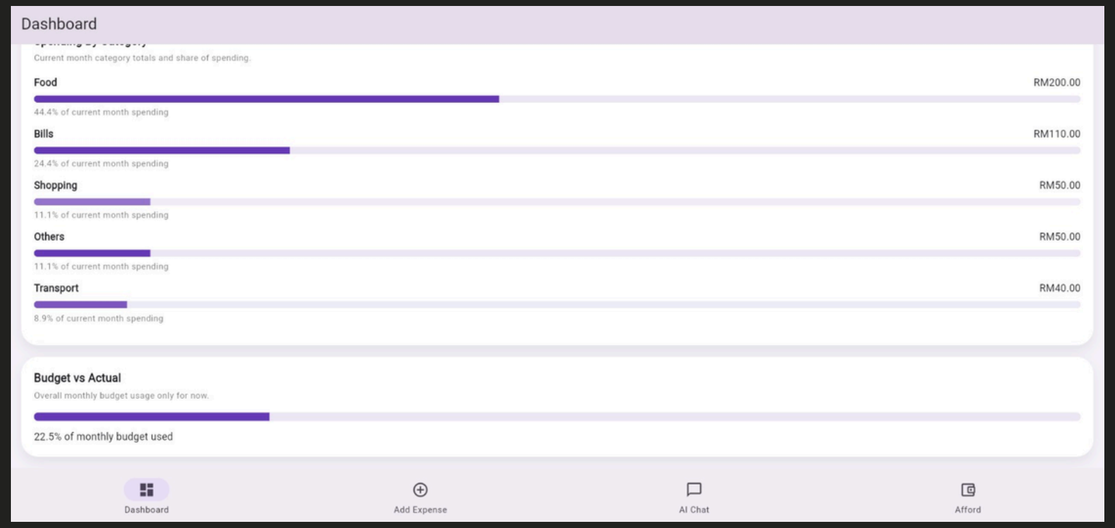
Features

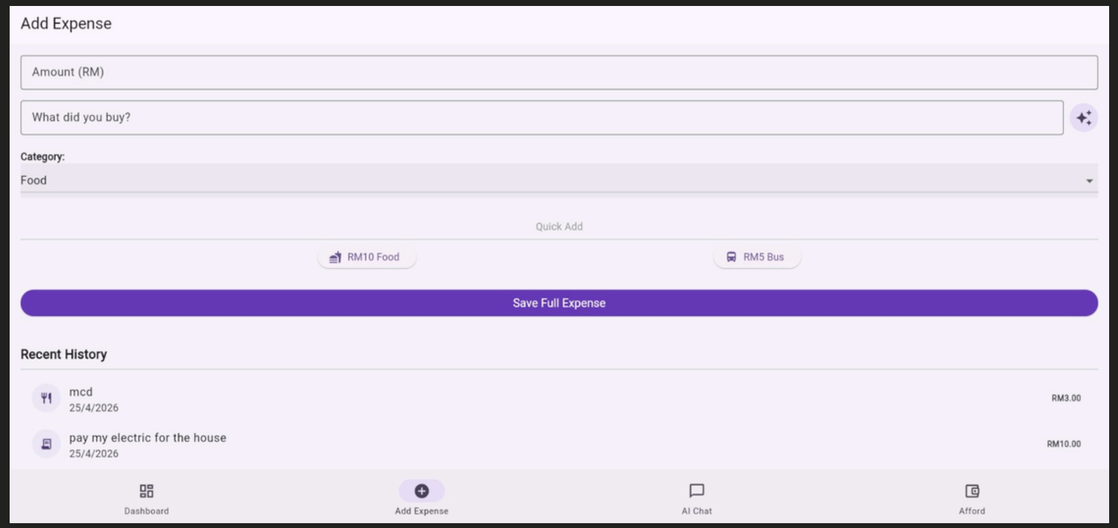
- Dashboard
- Add Expense
- AI Chat
- Affordability Checker

DASHBOARD + ADD EXPENSE

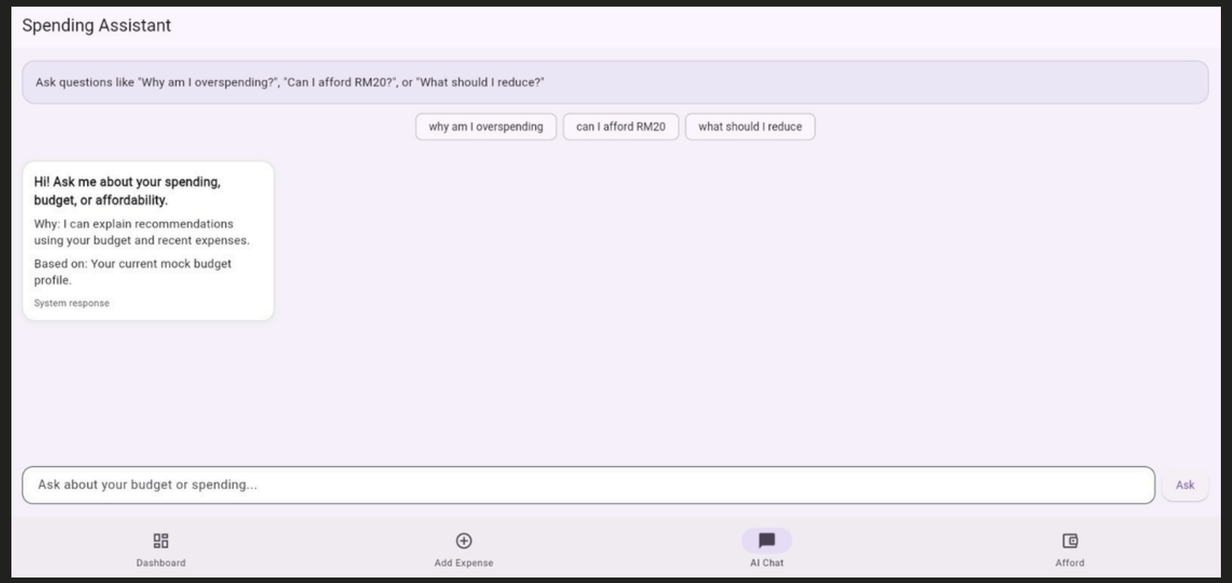


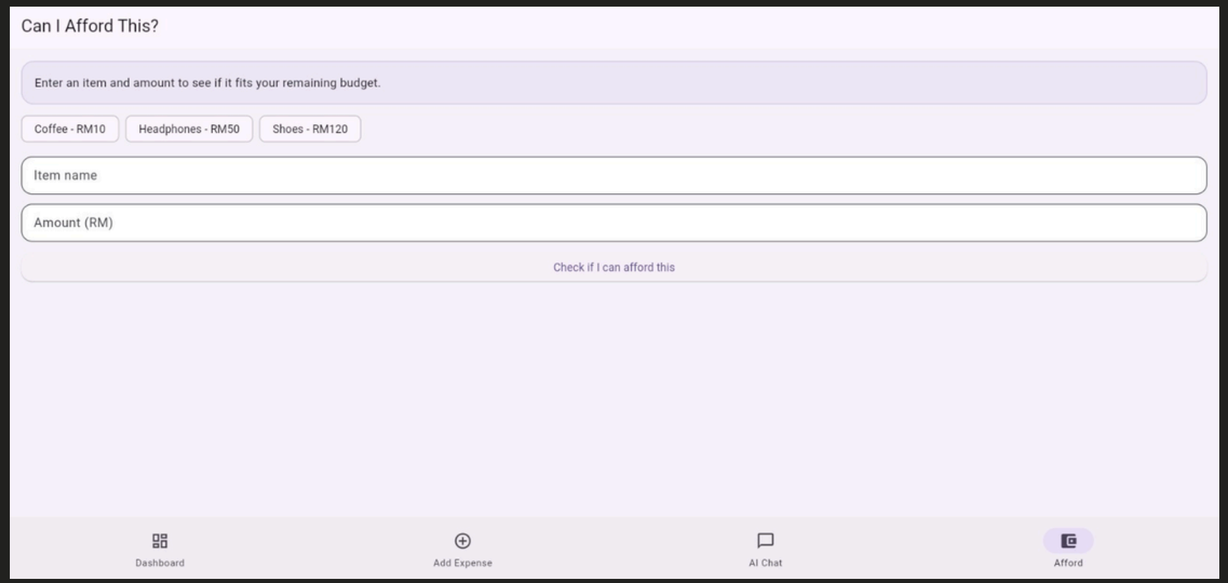




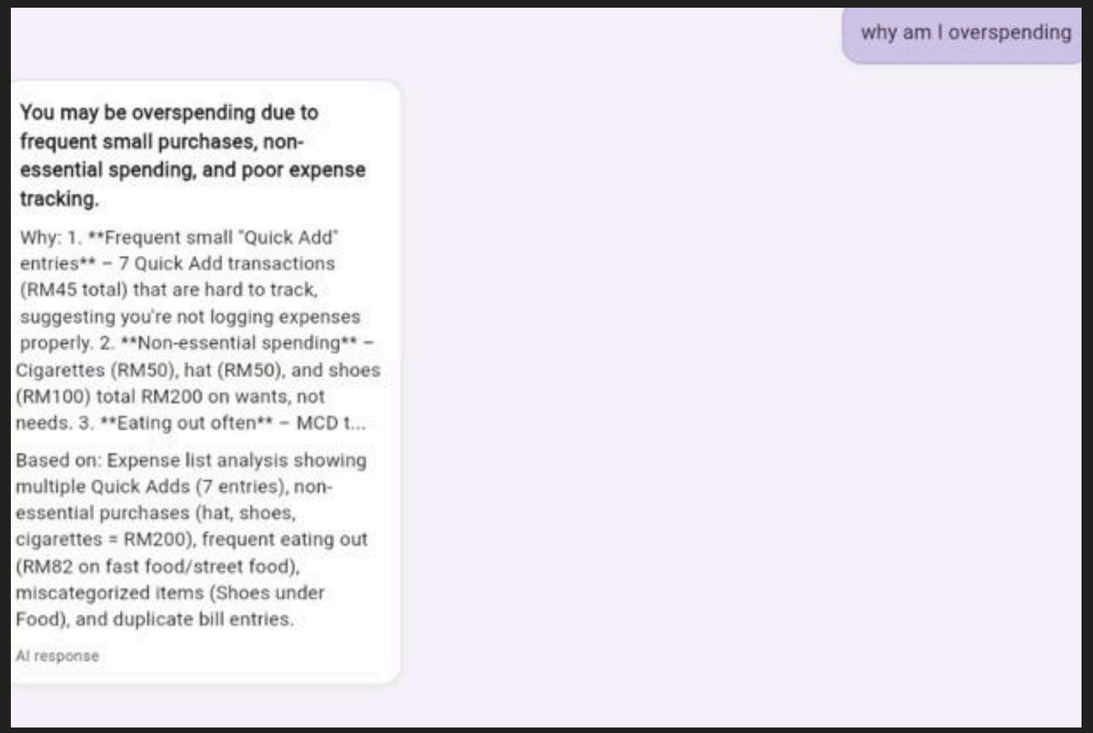


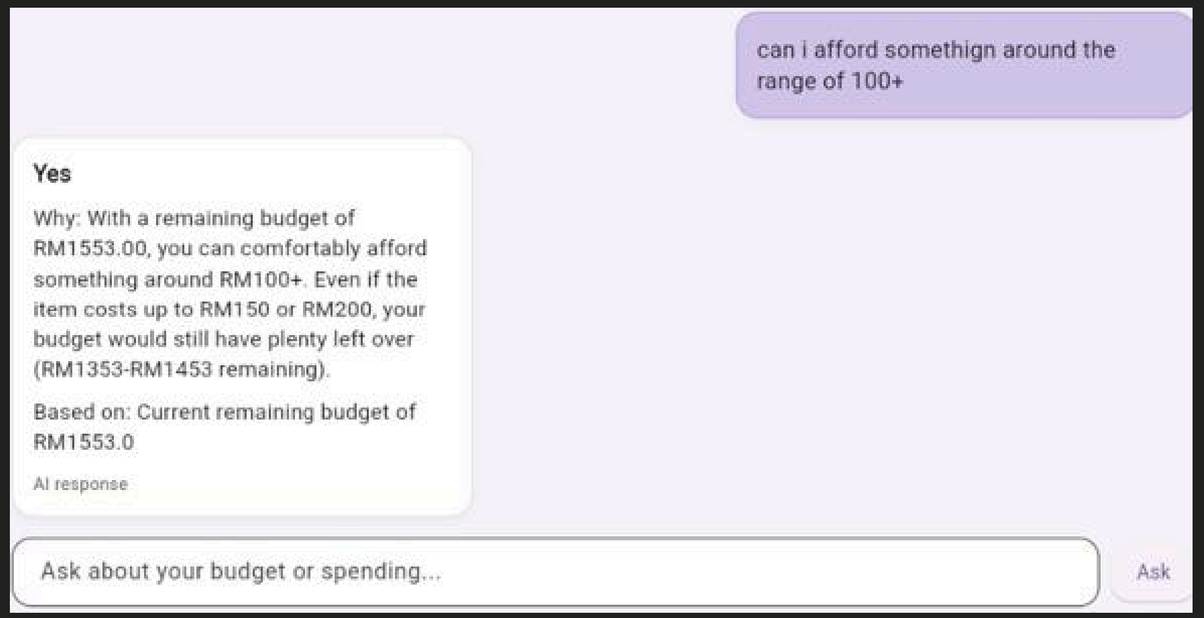
AI CHAT + AFFORDABILITY CHECKER





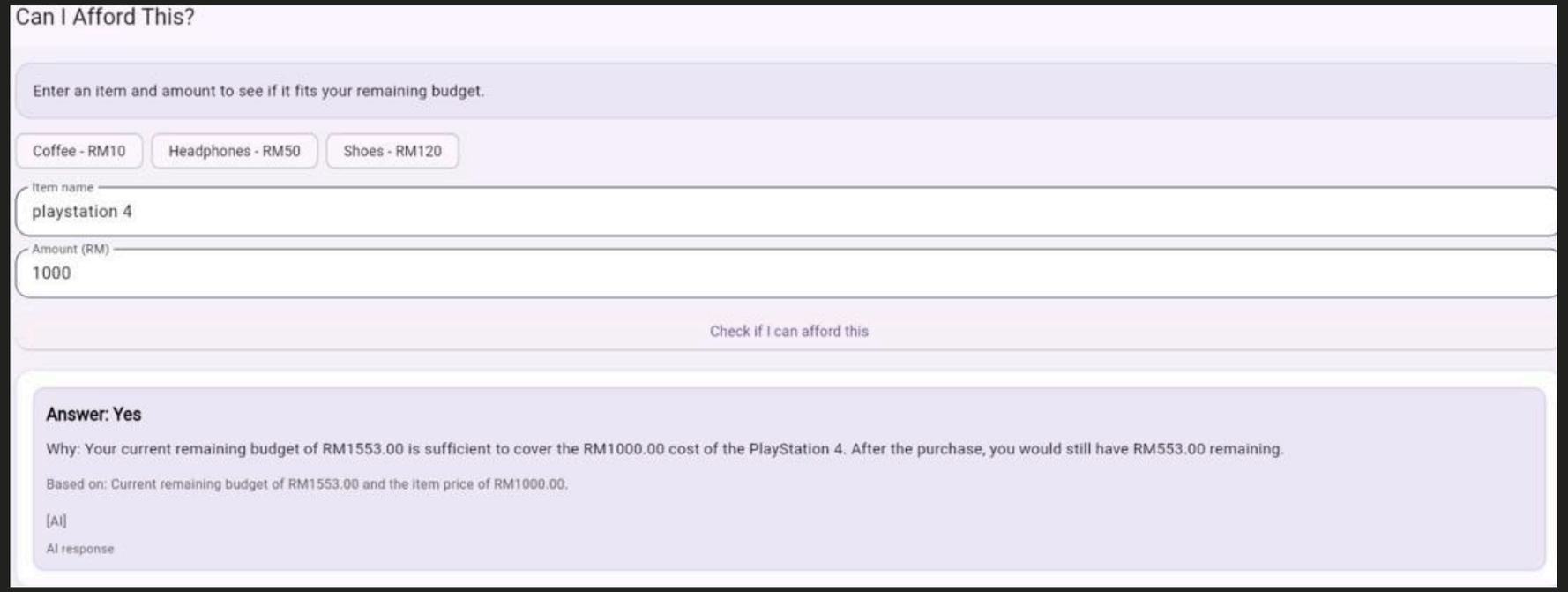
EXAMPLE: AI CHAT





EXAMPLE: AFFORDABILITY CHECKER

EXAMPLE: AFFORDABILITY CHECKER



QUANTIFIABLE IMPACT

Before (No app)	After (With app)
Runs out of money by week 3	Has money left at month end
Guesses “can I afford this?” (60% wrong)	Knows for sure (90% correct)
Stresses about spending	Confident and in control

Result: Potential to save up to RM200+ per month

TECHNICAL ARCHITECTURE

- App (Flutter) - Shows dashboard, captures expenses
- Cloud Database (Firestore) - Stores spending records
- Backend (FastAPI) - Connects app to AI, handles fallback
- Z.AI GLM (The Brain) - Handles affordability checker and chat questions

What happens when you ask a question?

1. App sends your expenses + questions to the backend
2. Backend calls Z.AI GLM
3. GLM returns Answer + Reason + Based on
4. App shows it to you

GLM INTEGRATION DETAILS

MODEL	Z.AI GLM – called via FastAPI backend
PROMPTING STRATEGY	Zero-shot + structured output: "Return JSON with Answer, Reason, Based on fields"
DATA SENT TO GLM	Recent expenses (recent history)
FALLBACK	Local deterministic rules (affordability, categorisation) when GLM unavailable
TOKEN COST ESTIMATE	~1,000 tokens per query → ~RM0.02 per user interaction

GLM DEPENDENCY VALIDATION

	WITH GLM	WITHOUT GLM (FALLBACK ONLY)
AFFORDABILITY REASONING	Context-aware: considers recent expenses and remaining budget	Simple: "RM250 > RM300 left = No"
SPENDING PATTERN INSIGHT	"Your recent expenses show higher spending in certain categories"	Limited ability to detect spending trends
EXPLANATION QUALITY	Plain language + specific examples	Rule-based: limited explanation based on budget values
DECISION INTELLIGENCE	Yes (provides actionable suggestions)	No (just shows numbers)

THANK YOU.